

Malio OCR benchmark — honest ranking

Generated 2026-10-07T20:35:39+00:00 UTC · 4 adjudicated phone-photo invoices · 7 engine configs scored · 1 BLOCKED (missing license/SDK).

What this ranks: per-field exact-match correctness, using the shared regex extractor from pdf-tests/extractor.py. The engine with the most correct fields across the manifest wins. Ties are listed. Fields marked UNREADABLE or AMBIGUOUS in the ground truth are excluded from denominators.

n = 4. Wilson 95% CIs span ~45 percentage points. This ranking is informative, not statistically conclusive. Roadmap §6 asks for ~100 documents to screen engineering decisions. Confidence intervals narrow materially only above n ≈ 50.

Ranking — all scorable fields

Counts both PRESENT fields (engine must extract the correct value) and ABSENT fields (engine must return null). Scoring 3/4 on allocation number mostly reflects correctly returning null on 3 docs where no allocation exists — see the next table for a stricter view.

Rank	Engine	Correct / Scorable	Accuracy	Wilson 95% CI	Docs won
1	google-vision-document-text-detection-CLOUD-CONTROLL	15 / 30	50.0%	[33.2%, 66.8%]	2.0
2	kraken-medium-cpu	10 / 30	33.3%	[19.2%, 51.2%]	0.3
3	paddleocr-hebrew-server-svtrv2-line	8 / 30	26.7%	[14.2%, 44.4%]	0.3
	paddleocr-hebrew-server-svtrv2-word	8 / 30	26.7%	[14.2%, 44.4%]	0.3
5	ironocr-heb-ara-eng	7 / 30	23.3%	[11.8%, 40.9%]	1.0
6	tesseract-best-heb-eng-psm3	6 / 30	20.0%	[9.5%, 37.3%]	0.0
	tesseract-fast-heb-eng-psm3	6 / 30	20.0%	[9.5%, 37.3%]	0.0
—	abbyy-fre12-heb-ara-eng	BLOCKED	n/a	—	—

Honest view — PRESENT fields only (did the engine actually extract?)

Removes the inflation from easy ABSENT matches. This is the harder question: when a field IS on the document, does the engine recover the correct value?

Rank	Engine	Correct / Scorable (PRESENT)	Accuracy	Wilson 95% CI
1	google-vision-document-text-detection-CLOUD-CONTROLL	12 / 27	44.4%	[27.6%, 62.7%]
2	kraken-medium-cpu	7 / 27	25.9%	[13.2%, 44.7%]
3	paddleocr-hebrew-server-svtrv2-line	5 / 27	18.5%	[8.2%, 36.7%]
	paddleocr-hebrew-server-svtrv2-word	5 / 27	18.5%	[8.2%, 36.7%]
5	ironocr-heb-ara-eng	4 / 27	14.8%	[5.9%, 32.5%]
6	tesseract-best-heb-eng-psm3	3 / 27	11.1%	[3.9%, 28.1%]
	tesseract-fast-heb-eng-psm3	3 / 27	11.1%	[3.9%, 28.1%]
—	abbyy-fre12-heb-ara-eng	BLOCKED	n/a	—

Per-field correctness matrix

One cell per (engine × field). Shows correct/scorable where scorable > 0, otherwise '—'. Fields excluded from this document's ground truth (UNREADABLE/AMBIGUOUS) do not count.

Engine	Vendor	Tax ID	Invoice#	Date	BeforeVAT	VAT	Total	Alloc#
google-vision-document-text-detection-CLOUD-CONTROL	1/3	1/4	2/4	2/3	2/4	2/4	2/4	3/4
kraken-medium-cpu	0/3	3/4	3/4	1/3	0/4	0/4	0/4	3/4
paddleocr-hebrew-server-svtrv2-line	0/3	2/4	2/4	1/3	0/4	0/4	0/4	3/4
paddleocr-hebrew-server-svtrv2-word	0/3	2/4	2/4	1/3	0/4	0/4	0/4	3/4
ironocr-heb-ara-eng	0/3	3/4	1/4	0/3	0/4	0/4	0/4	3/4
tesseract-best-heb-eng-psm3	1/3	2/4	0/4	0/3	0/4	0/4	0/4	3/4
tesseract-fast-heb-eng-psm3	1/3	2/4	0/4	0/3	0/4	0/4	0/4	3/4
abbyy-fre12-heb-ara-eng	BLKD	BLKD	BLKD	BLKD	BLKD	BLKD	BLKD	BLKD

Per-document winner (critical-fields-correct)

Which engine extracted the most fields correctly on each document. Ties shared.

Document	Winning engine(s)
Handwritten	kraken-medium-cpu paddleocr-hebrew-server-svtrv2-line paddleocr-hebrew-server-svtrv2-word
photo_2_2	ironocr-heb-ara-eng
photo_3_1	google-vision-document-text-detection-CLOUD-CONTROL
photo_4	google-vision-document-text-detection-CLOUD-CONTROL

Methodology

Candidates (local — six per roadmap §4): RivoksLab/paddleocr-hebrew (word, line), Tesseract + tessdata_best, Tesseract + tessdata_fast, Kraken (medium multilingual), IronOCR (2026.10.2 + Hebrew/Arabic language packs), ABBYY FineReader Engine 12.

Reference control: Google Cloud Vision DOCUMENT_TEXT_DETECTION, included as a CLOUD control (not one of the six candidates — roadmap §1 forbids hosted OCR in the main benchmark). Rendered in the same tables for direct comparison, labeled with -CLOUD-CONTROL suffix.

Scoring rules (roadmap §7): exact-match per field after NFC + whitespace normalization. IDs are strings (leading zeros preserved); amounts compared via Python Decimal at annotated precision; dates compared as ISO yyyy-mm-dd. No edit-distance tolerance in primary scoring.

Downstream parser: all engines pipe through the same deterministic regex extractor (pdf-tests/extractor.py). The parser is the bottleneck for every engine — its label patterns were built for text-layer PDFs, so Hebrew OCR noise (n↔n, n↔o substitutions, dot-vs-comma decimals) breaks label matching even when the raw OCR has the right tokens.

Ground truth: adjudicated by Claude reading each image directly (2026-10-07). Vendor name convention follows roadmap §2 (= the דוֹבָנֵל/counterparty on the invoice). The handwritten receipt's invoice_date and the Migdal invoice's truncated vendor_name are marked AMBIGUOUS and excluded from their denominators.

What this ranking does NOT measure

- Latency, RAM, GPU, cost per 1M documents, cost per correct invoice — all roadmap §9 columns. Would require workload measurements + dated license quotes.
- Raw-text OCR quality — e.g., Google Vision captures amounts in raw text that the parser then fails to extract. Judged on the final canonical JSON, not the OCR substrate.
- Ensemble/routing performance — roadmap §5 options A/B/C routers are not yet simulated.
- Script-specific accuracy — e.g., Arabic-only subsets. n=4 is too small.
- Statistical ranking — Wilson CIs are too wide at n=4 to establish "engine X is better than engine Y" with confidence.